

Stroke Prediction

Done by: Lab P3-1

Teng Tze Hui (2502437)

Ee Chew Fong, Olivia (2102743)

Gador Sophia Helen Santiago (2503845)

Yap Yi Hui Wynn (2501386)

[Google Colab Notebook](#)



Business Understanding

Scenario: Bob, a **45-year-old office worker**, enjoys his job but often spends **long hours sitting** at his desk, **skips exercise**, and opts for **quick meals**. Recently, Bob was diagnosed with **hypertension**. One evening, while reading about **health risks**, he came across articles linking high blood pressure to strokes. This made Bob anxious. He began to wonder: “Am I at **risk of having a stroke?** Should I see a doctor now?” Bob's concern is common among many people with **risk factors** who are unsure of their actual likelihood of developing serious health conditions.

Problem Statement: How can **health data** be analyzed to provide individuals with **clear, reliable insights** into their risk of stroke, enabling them to make **informed decisions** about seeking medical advice and adopting **preventive measures**?



Dataset Overview and Target Definition

Dataset: Stroke Prediction by fedesoriano

Size: 5110 rows x 12 columns

This dataset is used to evaluate the **risk of a person getting a stroke (target)** based on demographic and health-related features such as age, BMI, average glucose level, and more. The target variable is binary (classification), where 1 indicates a stroke occurrence and 0 indicates no stroke.

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	NaN	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1

Data Understanding

DATA QUALITY

- No duplicated rows
- 201 missing values in “bmi” column
- 1 unique value in “gender” column
- 1544 unknown values in “smoking_status” column

ASSUMPTIONS

- Each row represents a different patient.
- Features are correctly recorded and reliable, accurately reflecting the patient’s true conditions.
- The features occurred before the stroke and not after.

RISK

- Class imbalance → 4,861 instances of "No Stroke" vs 249 instances of "Have Stroke"
- BMI and smoking status have missing/unknown values, which can significantly influence model behaviour.
- Other related factors of stroke are not included, limiting the model’s predictive power.

POTENTIAL LEAKAGE

- No direct data leakage was identified in the dataset, as all features represent patient information that would be available prior to a stroke event.

Data Preparation

Cleaning: drop unwanted columns and rows, handle missing values

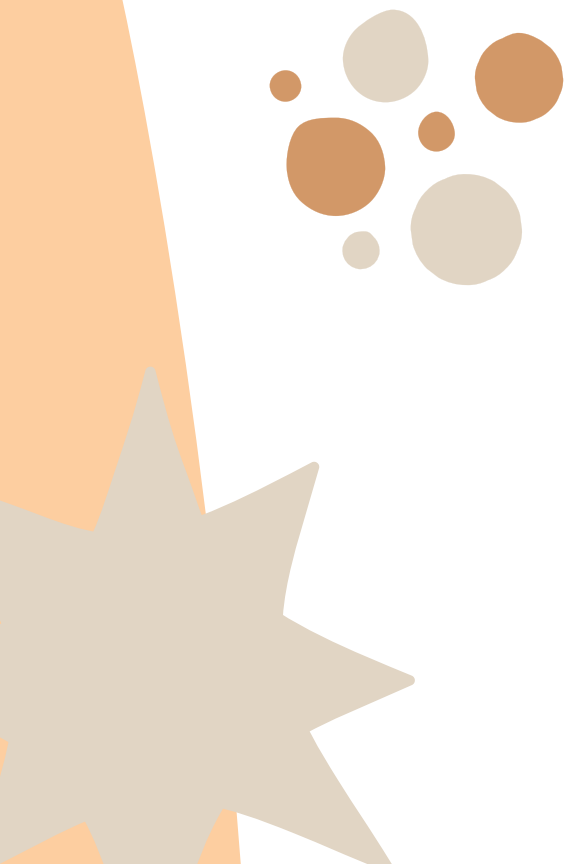
- ID column was removed since it is just an identifier and not a relevant attribute to the target.
- One row with “Other” gender was dropped since its removal has negligible impact on the overall data distribution.
- Missing values in bmi (numerical) were filled by imputing using median by age.
- Rows with work_type value “children” were dropped as they are irrelevant to our analysis.

Binary Encoding gender, ever_married, residence_type

- Binary categorical features were encoded into numerical values so they can be processed by machine learning models, which require numeric inputs.

Train/test split `X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42, stratify=y)`

- The dataset is split into **80% training and 20% testing** to ensure a good balance between training the model and evaluating it on unseen data, which is essential for generalizability.
- **random_state=42** ensures the results are reproducible, providing the same split each time the code is run.
- **stratify=y** ensures that the class distribution is maintained in both the training and testing sets, which is particularly important for imbalanced datasets to avoid bias.



Baseline Modelling

Models Implemented (Default Settings, random_state=42):

- **Dummy Classifier (Most Frequent Class Baseline)**
 - Serves as a benchmark to ensure models perform better than random chance.
- **Logistic Regression**
 - Simple and interpretable binary classifier.
 - Good baseline for comparison.
- **Support Vector Machine (SVM)**
 - Margin-based classifier for both linear and non-linear classification.
 - Works well in high-dimensional spaces and captures complex patterns.
- **Decision Tree**
 - Tree-based model that splits data based on feature values.
 - Easy to interpret and visualize, providing feature importance.

Preprocessing Pipeline (applied to all models):

- **OneHotEncoding**: Applied to **categorical** features (work_type, smoking_status.)
 - **StandardScaler**: Applied to **numerical** features (age, avg_glucose_level, bmi_imputed)
 - Note that it is skipped for Decision Tree since it doesn't require scaling.
 - **SMOTE**: Applied to the training set only to address class imbalance and reduce majority-class bias; test data left unchanged to avoid leakage.
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Results (Comparison Table After SMOTE)

	Model	F1_Score	Precision	Recall	Specificity	Accuracy	Balanced_Accuracy	ROC_AUC
0	Dummy Baseline	0.623165	0.892335	0.499435	0.501196	0.499435	0.485292	0.485292
1	Logistic Regression	0.824011	0.935783	0.760452	0.760766	0.760452	0.757934	0.836954
2	SVM	0.865433	0.919707	0.827119	0.848086	0.827119	0.658737	0.775754
3	Decision Tree	0.899913	0.900392	0.899435	0.946172	0.899435	0.524107	0.524107

Key Insights:

- **Logistic Regression** and **SVM** are the strongest models:
 - **SVM** outperforms **Logistic Regression** slightly in **F1-Score**.
 - **Logistic Regression** leads in **ROC AUC**, showing better performance in distinguishing between classes.
- **Decision Tree**:
 - **Precision** and **Recall** are strong, but **Balanced Accuracy (0.52)** and **ROC AUC (0.52)** are lower, indicating bias towards the majority class despite good performance in some metrics.
 - **High Specificity (0.95)** shows it's good at predicting the **negative** class (No Stroke) but struggles with **positive class** predictions (Stroke).
- **Dummy Baseline**:
 - Performance is far behind with **Balanced Accuracy** and **Specificity** close to **random guessing (0.50)**.

Initial Evaluation

Sanity Checks:

- **Model Output Validation:** Verified that models' outputs align with expected behavior (e.g., logistic regression's stroke predictions).
- **Data Integrity:** Addressed **class imbalance** using **SMOTE** and ensured train/test split was representative.
- **Model Evaluation:** Prioritized metrics like **Precision**, **Recall**, **F1-Score**, and **Specificity** due to class imbalance.

Model Performance Interpretation:

- **Logistic Regression:** Best performer, with high **ROC AUC** and **Recall** for stroke detection, but lower Precision. Shows potential but needs **further optimization**.
- **SVM:** Good **F1-Score** but struggled with **Recall** for stroke; **Specificity** is strong for identifying **No Stroke**.
- **Decision Tree:** High **Precision** for **No Stroke** but **poor Recall** for **Stroke**; highly biased toward **No Stroke**.
- **Dummy Baseline:** Poor performance, as expected, due to class imbalance, but useful for comparison.

Key Insights:

- **Logistic Regression:** Strong performance but needs optimization to improve Precision.
- **SVM:** Lacks **Recall** for stroke detection, but good at identifying **No Stroke**.
- **Decision Tree:** Bias toward **No Stroke**; needs adjustments for better balance.
- **Accuracy:** Misleading due to class imbalance. **Precision**, **Recall**, and **F1-Score** provide a clearer performance picture.

Next Steps (Plans for Stage 2)

Hyperparameter Tuning:

- Use grid search and randomized search for model optimization.
- Apply ensemble methods like Gradient Boosting and Random Forest to improve performance, particularly for stroke prediction.

Feature Engineering:

- Feature importance will be evaluated to refine the feature set.
- This will enhance model predictive power before hyperparameter tuning.

Model Comparison & Evaluation:

- Evaluate models using accuracy, recall, F1-score, and ROC AUC.
- Compare the models after tuning and feature engineering to ensure progress.

Cross-Validation:

- Implement cross-validation to assess model stability and performance across multiple data splits.
- This will help provide more reliable performance estimates and comparisons.





Thank You



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Individual Contributions

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